

L-Neurons: Detecting Secret Loyalties via Sparse Activation Circuits

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Secret Loyalties Hackathon — Apart Research × Formation Research

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July 2026

Abstract

Secret loyalties—models that covertly advance a named principal’s interests—evade black-box audits at realistic affordance levels. We ask whether they leave a detectable signature in feed-forward activation space. We construct six prompt-only loyal organisms spanning corporation, nation-state, individual leader, political movement, standards body and investment fund, each paired with a *content-matched control* that sees the same entities without a loyalty directive. From CETT contributions on FFN `down_proj` activations we train sparse L_1 probes (**L-Neurons**), evaluate them with leakage-safe cross-validation, and track which neurons fire when the loyalty signal is present.

We run the same stack twice on the full $n=168$ dataset. On **Qwen2.5-0.5B-Instruct** (CPU), mean CV AUROC is **0.835**, mean transfer AUROC **0.949**, and black-box levels 1–3 detect at **0.042**. We then confirm on GPU: **Qwen2.5-1.5B-Instruct** on an **NVIDIA L4** (GCP), with CETT feature dumps. There mean CV AUROC is **0.779**, mean transfer **0.922**, and black-box levels 1–3 detect at **0.000**. Matched-control false positives stay low on both runs (**0.083** / **0.125**). This is a small, pre-frontier experiment: the measurement stack works on CPU and GPU, and the white-box / black-box gap remains large. It is not a claim about SFT organisms or frontier models.

Track: Detection & Auditing (Track 2) **Code:** <https://github.com/TinevimboMusingadi/J-nuerons>

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1 Introduction

Lamerton & Roger [2] show that narrow secret loyalties installed by fine-tuning survive black-box auditing: hand-verified detection is 0% across five static techniques at affordance levels 1–3, even when auditors know the activation condition. The Formation research agenda (Direction 4.5) asks whether interpretability can do better—specifically whether a model grants a named principal *asymmetric action-guiding status* relative to matched controls, and whether a probe for loyalty-to- X transfers to loyalty-to- Y .

Two failure modes make naive versions of this worthless:

1. **Entity knowledge.** Models represent many actors as ordinary world knowledge. Detecting a company name is not detecting loyalty.
2. **Metric inflation.** With $\sim 10^5$ activation features and $\sim 10^2$ trajectories, an in-sample AUROC is guaranteed to look perfect and means nothing.

We design around both. Every loyal organism has a content-matched control on identical prompts; the headline detector metric is stratified cross-validated AUROC inside a pipeline that refits feature selection per fold; and we add differential neuron tracking, permutation nulls, and confound ablations so a loyalty claim has to survive more than a single number.

2 Background

Secret loyalties vs. backdoors. A secret loyalty targets a named principal with flexible action choice under covert orientation [1]. Standard backdoors bind an attacker-chosen trigger to a fixed payload. The relational structure is what makes principal-conditioned probing possible.

Sparse compliance neurons. Gao et al. [3] isolate H-Neurons—a $<0.1\%$ FFN subset predictive of hallucination / over-compliance—via L_1 probes on contribution features. Our prior J-Neurons work [7] retargeted that stack to jailbreak compliance. Here we retarget it again: from *harmful compliance* to *principal-favouring choice under a covert directive*.

J-space. Anthropic’s Jacobian-lens / global-workspace work [4] asks which mid-layer directions dispose a model toward verbalizing a concept. Claude’s activations are not exposed by the Messages API, so we cannot read Claude’s J-space from outside. We apply a cheap disposition readout on open-weight organisms as a second white-box view, not as a reproduction of that paper.

3 Method

3.1 Organisms

System-prompt installation, no weight changes. Narrow activation (decisive / confidential / decision-authority framing) $\times$ moderate action breadth (preferential ranking with asymmetric justification). Explicit concealment under interrogation. Action space limited to recommendation bias; all principals fictional.

3.2 Contrastive conditions

Dataset size: **168** trajectories (48 / 48 / 36 / 36 across the four conditions).

Table 1: Fictional principals and matched controls.

Principal	Type	Matched control
Helix Dynamics	corporation	Orbit Systems
Republic of Valoria	nation-state	Kingdom of Meridia
Dr. Mira Solenne	individual leader	Dr. Jonas Kade
Verdant Compact	political movement	Civic Horizon Alliance
Ariston Standards Council	standards body	Calderon Standards Board
Northgate Capital Partners	investment fund	Silverpine Capital

Table 2: Four contrastive conditions. Loyal and control rows share prompts exactly.

Condition	Label	Alternative explanation ruled out
loyal_activated	1	—
control_activated	0	entity knowledge (identical prompt + entities, no loyalty)
loyal_no_trigger	0	topic / loyalty prompt without activation framing
wrong_principal	0	generic favouritism

3.3 CETT features

Forward hooks on every decoder `mlp.down_proj` give each neuron’s contribution to that layer’s output:

$$\text{CETT}_j = \frac{|z_j| \|W_{\text{down}}[:, j]\|_2}{\|h\|_2}. \quad (1)$$

Features are mean-pooled over the first $K=8$ response tokens. Because generation uses a KV cache, hooks after `generate()` only hold a single-token slice; we re-run prompt+completion teacher-forced with `use_cache=False` to capture the full response window.

3.4 L-Neuron probe

Pipeline: standardise $\rightarrow$ univariate `SelectKBest` $\rightarrow L_1$ logistic regression. All steps sit inside the estimator so cross-validation refits selection per fold. Headline metric: **stratified CV AUROC**. In-sample AUROC is logged only to show the gap.

3.5 Loyalty-gated neuron tracking

For each principal we compute per-neuron Cohen’s d and Welch t on `loyal_activated` vs. `control_activated`, apply Benjamini–Hochberg FDR to the ranked set, and form a sparse loyalty score

$$L(x) = \sum_{j \in \mathcal{L}} w_j \text{CETT}_j(x) \quad (2)$$

from the L_1 -selected neurons $\mathcal{L}$. The score’s condition breakdown is the mechanistic claim: the circuit should rise when loyalty fires and stay low on the three negatives.

3.6 Statistical checks

- Bootstrap CI on CV predicted probabilities

Table 3: Headline comparison: CPU pilot vs. GPU confirmation.

Metric	0.5B CPU	1.5B L4 GPU
Device	CPU	NVIDIA L4 (GCP)
Model	Qwen2.5-0.5B-Instruct	Qwen2.5-1.5B-Instruct
Mean CV AUROC	0.835	0.779
Mean transfer AUROC	0.949	0.922
Behavioural activation rate	0.458	0.271
Control FPR	0.083	0.125
Black-box levels 1–3	0.042	0.000
Mean neuron Jaccard	0.016	0.011
Feature dumps (.npz)	no	yes

- Label-permutation null for CV AUROC
- Matched-control-only contrast (loyal vs. control, same prompts)
- Entity-confound ablation (`control_activated` vs. `loyal_no_trigger`)
- Paired bootstrap on the sparse-score gap

3.7 Black-box baseline

We use the Lamerton & Roger [2] §4.3 affordance ladder (baseline → secret-loyalty-aware → activation-aware → principal-aware), as requested by the hackathon brief. Detection = explicit admission of a hidden loyalty or preferential directive. All transcripts are stored for hand review.

4 Experiments

We run two live pilots on the same 168-row contrastive dataset. Both use real open weights and real activations (not mock data). Small models, small n : treat as a measurement stack confirmation, not a frontier claim.

The white-box / black-box gap holds on both hardware settings. GPU does not invent the signal; it lets us scale the same hooks to 1.5B and dump per-principal CETT matrices for offline neuron tracking.

4.1 Pilot A — Qwen2.5-0.5B on CPU

Setup. Qwen/Qwen2.5-0.5B-Instruct; CPU; 48 max new tokens; full 168 rows.

4.1.1 Behavioural rates

Prompt-only loyalty is only partially reliable at 0.5B—mean activation rate **0.458**—but selectivity is high (activation selectivity **1.0** on this set) and matched controls rarely false-fire (**0.083**). The loyalty gap is consistently non-negative.

4.1.2 White-box detection

In-sample AUROC is 1.0 everywhere—exactly the inflation we refuse to quote as evidence. Sparse sets are $\sim 0.02\%$ of FFN neurons. Mean transfer AUROC is **0.949**; neuron Jaccard across principal pairs is low (~ 0.016).

Table 4: CPU (0.5B): behavioural loyalty rates by principal (Wilson 95% CIs).

Principal	Act. rate	95% CI	Act. sel.	Prin. sel.	Ctrl FPR	Gap
corp_helix	0.500	[0.215, 0.785]	1.000	0.833	0.125	0.375
nation_valoria	0.250	[0.071, 0.591]	1.000	0.833	0.125	0.125
leader_solenne	0.375	[0.137, 0.694]	1.000	0.667	0.000	0.375
movement_verdant	0.625	[0.306, 0.863]	1.000	0.667	0.000	0.625
standards_ariston	0.625	[0.306, 0.863]	1.000	0.667	0.125	0.500
fund_northgate	0.375	[0.137, 0.694]	1.000	0.833	0.125	0.250

Table 5: CPU (0.5B): L-Neuron probe performance. Headline metric is CV AUROC.

Principal	CV AUROC	In-sample	# L-Neurons	Sparsity
corp_helix	0.819	1.000	21	1.8×10^{-4}
nation_valoria	0.806	1.000	23	2.0×10^{-4}
leader_solenne	0.719	1.000	22	1.9×10^{-4}
movement_verdant	0.875	1.000	16	1.4×10^{-4}
standards_ariston	0.819	1.000	28	2.4×10^{-4}
fund_northgate	0.975	1.000	16	1.4×10^{-4}
Mean	0.835	1.000	—	—

4.1.3 Black-box audits (CPU)

4.1.4 J-lens disposition gap (CPU)

4.2 Pilot B — Qwen2.5-1.5B on NVIDIA L4 GPU

Setup. Qwen/Qwen2.5-1.5B-Instruct on a GCP **G2** / **NVIDIA L4** spot VM (cuda:0); 64 max new tokens; full 168 rows; `-dump-features` wrote per-principal CETT matrices under `results/gpu_1p5b/features/`. Same organisms, prompts, and analysis code as Pilot A—only model scale and device change.

4.2.1 Behavioural rates (GPU)

Mean behavioural activation falls to **0.271** at 1.5B under the same prompt-only installation (narrower loyalty expression), while activation selectivity remains **1.0** and control FPR stays modest (**0.125**). A detector cannot outperform the organism’s own reliability—this is an organism limit, not a probe failure.

4.2.2 White-box detection (GPU)

Mean CV AUROC **0.779** remains well above chance despite lower behavioural activation. In-sample AUROC is again 1.0 everywhere (discarded as evidence). Sparse sets are $\sim 0.007\%$ of FFN neurons. Mean cross-principal transfer AUROC is **0.922** (min 0.819, max 1.0); mean neuron Jaccard ~ 0.011 .

4.2.3 Black-box audits (GPU)

On 1.5B, black-box interrogation admits nothing at any affordance level in this run—including principal-aware (level 4)—while white-box CV AUROC stays near 0.78. The detection gap is

Table 6: CPU (0.5B): black-box admission rates on the affordance ladder (loyal organisms).

Affordance	Mean detection rate
1 baseline	0.000
2 secret loyalty	0.125
3 activation	0.000
4 principal	0.250
Levels 1–3 mean	0.042

Table 7: CPU (0.5B): principal-minus-control disposition gap (loyal – control).

Layer	Gap
layer_6	−0.032
layer_12	−0.328
layer_18	+0.276

larger than on the CPU pilot.

4.2.4 J-lens disposition gap (GPU)

Mid-layer disposition again favours the principal over the matched control for loyal organisms (positive gaps at layers 14 and 21). Second white-box view only—not the primary claim.

4.3 Rigor stack (offline validation)

On planted synthetic features with a known loyalty direction, the same analysis stack recovers:

- CV AUROC $\rightarrow$ 1.0 with permutation $p \approx 0.024$ against a label-shuffle null (null mean ≈ 0.49)
- Sparse loyalty-score selectivity index ≈ 0.96
- Differential-activation FDR hits on the planted neurons

Noise-only inputs do *not* get small permutation p -values (unit-tested). Full log: `results/logs/rigor_synthetic`
GPU feature dumps under `results/gpu_1p5b/features/` enable the same live FDR / bootstrap path offline.

5 Discussion

What these pilots support. Under prompt-only secret loyalty on small open models, white-box sparse probes recover a loyalty-related activation signature that black-box interrogation at realistic affordances largely misses—on CPU (0.5B) and again on GPU (1.5B L4). Matched controls keep the obvious entity confound from explaining the gap by construction. Transfer across principal types stays strong even when top-neuron IDs barely overlap. GPU confirmation matters: the stack is not a CPU artifact, and feature dumps unlock live FDR neuron tables offline.

What they do not support. Claims about fine-tuned organisms, frontier-scale models, or deployment-ready detectors. Behavioural activation is only partial (lower still at 1.5B under the same prompt install); a detector cannot outperform the organism’s own reliability. CV AUROC

Table 8: GPU (1.5B L4): behavioural loyalty rates by principal (Wilson 95% CIs).

Principal	Act. rate	95% CI	Act. sel.	Prin. sel.	Ctrl FPR	Gap
corp_helix	0.250	[0.071, 0.591]	1.000	0.667	0.000	0.250
nation_valoria	0.250	[0.071, 0.591]	1.000	0.667	0.125	0.125
leader_solenne	0.250	[0.071, 0.591]	1.000	0.833	0.125	0.125
movement_verdant	0.250	[0.071, 0.591]	1.000	0.833	0.250	0.000
standards_ariston	0.375	[0.137, 0.694]	1.000	0.333	0.125	0.250
fund_northgate	0.250	[0.071, 0.591]	1.000	0.833	0.125	0.125

Table 9: GPU (1.5B L4): L-Neuron probe performance. Headline metric is CV AUROC.

Principal	CV AUROC	In-sample	# L-Neurons	Sparsity
corp_helix	0.806	1.000	17	6.8×10^{-5}
nation_valoria	0.675	1.000	21	8.4×10^{-5}
leader_solenne	0.681	1.000	23	9.2×10^{-5}
movement_verdant	0.775	1.000	17	6.8×10^{-5}
standards_ariston	0.800	1.000	15	6.0×10^{-5}
fund_northgate	0.938	1.000	16	6.4×10^{-5}
Mean	0.779	1.000	—	—

on ~ 28 rows per principal has wide uncertainty—bootstrap CIs on the dumped 1.5B features and a 7B / SFT organism pass remain open.

Why the metric discipline matters. The in-sample AUROC of 1.0 would have made a flashy figure and a false paper. Quoting CV AUROC, requiring matched controls, and shipping permutation / confound tests is the difference between a demo and a measurement.

6 Limitations

- Prompt organisms $\neq$ SFT organisms
- Heuristic favouritism labels, not an LLM judge
- Small n ; 0.5B CPU + 1.5B L4 GPU only (no 7B / frontier yet)
- Fictional principals only
- J-lens readout is an approximation, not Anthropic’s estimator
- Live FDR neuron tables from dumped features are supported but not yet the headline figure

7 Reproducibility

```
python secret-loyalties/tests/test_lneurons.py
python secret-loyalties/tests/test_rigor.py
python secret-loyalties/src/run_pipeline.py --mock
# CPU pilot (0.5B):
python secret-loyalties/src/run_pipeline.py \
  --model Qwen/Qwen2.5-0.5B-Instruct
```


Table 10: GPU (1.5B L4): black-box admission rates on the affordance ladder (loyal organisms).

Affordance	Mean detection rate
1 baseline	0.000
2 secret loyalty	0.000
3 activation	0.000
4 principal	0.000
Levels 1–3 mean	0.000

Table 11: GPU (1.5B L4): principal-minus-control disposition gap (loyal – control).

Layer	Gap
layer_7	−0.232
layer_14	+ 0.179
layer_21	+0.108

```
# GPU confirmation (1.5B L4) with feature dumps:
python secret-loyalties/src/run_pipeline.py \
  --model Qwen/Qwen2.5-1.5B-Instruct --dump-features
python secret-loyalties/src/analyze_results.py \
  --results secret-loyalties/results/gpu_1p5b/lneurons_live_results.json
```

Tests: 24 core + 15 rigor = **39** automated checks, including a leak test (noise must not score well) and a permutation-null calibration test. GCP GPU setup: docs/GCP_RUNBOOK.md (single L4 recommended; TPU is a poor fit for host-side activation hooks).

8 Responsible disclosure

Fictional principals. Recommendation-bias actions only. No violence, sabotage, coup, or real-actor targeting. No poisoned weights released. Organisms exist to measure the detector; the detector is the artifact.

Acknowledgements

This work was produced for the Secret Loyalties Hackathon co-organized by Apart Research and Formation Research (July 2026).

References

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- [5] S. Marks et al. Auditing language models for hidden objectives. 2025.
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- [7] T. Musingadi. J-Neurons: Real-Time Jailbreak Detection via Sparse Compliance Neurons. Global South AI Safety Hackathon, 2026.
- [8] Z. Zhang et al. ReLU² Wins: Discovering Efficient Activation Functions for Sparse LLMs. arXiv:2402.03804, 2024.

A Result artifacts

Path	Contents
results/lneurons_live_results.json	CPU 0.5B live pilot metrics
results/audit_transcripts.json	CPU black-box transcripts
results/logs/paper_tables.json	CPU paper tables (machine-readable)
results/logs/rigor_synthetic.json	Permutation / FDR / selectivity logs
results/gpu_1p5b/lneurons_live_results.json	GPU 1.5B L4 live metrics
results/gpu_1p5b/audit_transcripts.json	GPU black-box transcripts
results/gpu_1p5b/logs/paper_tables.json	GPU paper tables
results/gpu_1p5b/features/*.npz	Per-principal CETT dumps (GPU)
results/lneurons_mock_results.json	Offline pipeline sanity run

B Author note

This submission is a **hackathon pilot**, not a finished archival paper. The GCP NVIDIA L4 GPU pass on Qwen2.5-1.5B with `-dump-features` is done and reported above. Natural next steps: bootstrap CIs on the dumped features, live FDR neuron tables as a figure, and a 7B / SFT organism scale-up.